A.H.Thiviru Malitha Weerakoon

Electronics & Telecommunication Engineering Undergraduate

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My Portfolio

Professional Summary

Electronics and Telecommunication Engineering undergraduate passionate about embedded systems, robotics, computer vision, and IoT-based product development. Experienced in designing electronic systems, PCB development, microcontroller-based applications, CAD modelling, and rapid prototyping. Strong interest in transforming engineering concepts into practical solutions through innovation and teamwork.

Education

B.Sc. Engineering (Hons) in Electronics and Telecommunication Engineering 2023 – Present
University of Moratuwa, Sri Lanka
CGPA = 3.91/4.00
Relevant Areas:
Embedded Systems, Digital Electronics, Analog Electronics,
Computer Vision,Robotics, Communication Systems, Signal Processing

G.C.E A/L - Physical science 2020 – 2022
st joseph vaz college
Z score - 2.7561 — District Rank - 02 (Puttalam) — Island Rank - 59
Combined Mathematics - A
Physics - A
Chemistry - A

G.C.E. O/L Examination 2019
Ananda National College Chilaw
Score: 8 As and 1 C
I managed to obtain A passes in the core subjects, ICT, and Business Studies. I received a C pass in English Literature.

Technical Skills

Programming: Python, C, C++, Java, MATLAB, Verilog
Embedded Systems: Arduino, ESP32, Raspberry Pi, STM32, Microcontrollers, Sensors, IoT Systems
Electronics: Analog and Digital Circuit Design, PCB Design, Circuit Simulation, Prototyping, Soldering
Engineering Software: SolidWorks, Altium, Blender,Wireshark, LTspice
Computer Vision: OpenCV, Image Processing, Object Detection

Engineering Projects

Smart Water Meter (Product Development Project) 2026

- Designed and developed an IoT-enabled smart water meter prototype.
- Developed sensing mechanism, embedded control system, and data processing pipeline.
- Designed mechanical enclosure and considered product certification requirements.
- Currently working on accuracy validation and market-ready improvements.

Smart Dengue Patient Monitoring System (AI + IoT Healthcare Platform) 2026

- Developed a Raspberry Pi-based IoT healthcare system for automated dengue patient monitoring and digital record management.
- Integrated sensors and modules for real-time monitoring of vital parameters and patient health data.
- Developed a centralized database and web dashboard for data visualization and medical record management.
- Designed a scalable architecture for future hospital-wide patient monitoring applications.

Fully Analog Noise-Canceling Microphone Mar 2026 – Jun 2026

- Designed a fully analog noise-canceling microphone system using dual microphone inputs.
- Developed signal conditioning circuits and analyzed performance using LTspice simulations.
- Built and tested the prototype with breadboard implementation and hardware validation.
- Currently working towards custom PCB development.

Vision-Based Multi-Tasking Autonomous Robot 2026

- Developed an autonomous robot integrating computer vision and embedded control systems.
- Implemented OpenCV-based image processing for object detection and task execution.
- Integrated cameras, sensors, and motor control systems for autonomous navigation.
- Optimized hardware-software integration for reliable robotic operation.

MediPack Pro - AI Enabled Automated Medicine Packaging System 2026

- Led the development of an AI-enabled automated medicine packaging system for improving pharmacy efficiency.
- Designed dispensing mechanisms, embedded control systems, and sensor-based automation.
- Developed a scalable healthcare solution focusing on accuracy, reliability, and hygiene.
- Achieved Semi-Finalist position at the SLIoT Competition 2026.

UART Implementation in FPGA using Verilog Apr 2026 – May 2026

- Designed and implemented a full-duplex UART transceiver using Verilog RTL.
- Developed FSM-based transmitter and receiver modules with synchronization techniques.
- Implemented baud-rate generation for 115200 bps communication using a 50 MHz clock.
- Verified FPGA implementation through simulation and hardware testing.

Analog Sound Regulator Amplifier Sep 2025 – Dec 2025

- Designed an analog sound regulation amplifier for maintaining consistent audio output levels.
- Developed the complete analog circuit design and performed circuit analysis.
- Contributed to PCB assembly, soldering, testing, and hardware troubleshooting.

Autonomous Task-Based Robot 2025

- Developed an autonomous robot capable of color detection, line following, box picking, and ramp climbing.
- Implemented sensor integration, control algorithms, and mechanical design.
- Achieved an A grade in Robot Design and Competition module.

Automated Sound Amplifier 2025

- Led a four-member team to design and develop an automated audio amplifier.
- Responsible for circuit design, component selection, testing and documentation.
- Completed PCB prototyping and hardware validation.

Onging Projects

Real-Time Road Sign Detection Driver Alerting System

Automated Market Structure SMC Chart Analyzer

Fully Analog Heartbeat Detector Circuit Design

5-Arm Radial Mobile Robot

Verilog HDL Digital System Design Portfolio

Experience

Co-Founder – W3 Solutions(Unregistered Star-up) 2026 – Present

- Providing 3D printing and engineering prototyping services.
- Designed custom mechanical components and product enclosures.

Certifications

- Java Programming Master – Online / EVOTECH EDUCATION 2024

Leadership & Activities

Finance & Inventory Manager – SPARK Branch, Electronic Club, University of Moratuwa
2026 – Present

Spark Challenge Coordinator – SPARK Branch, Electronic Club, University of Moratuwa
2025 – 2026

- Active member involved in electronics workshops and engineering activities.
- Conducted Raspberry Pi workshops for school students.

Achievements

- Finalist – SLRC 2026
- Semi-Finalist – SLIoT Competition 2026
- 3rd Runner-Up – SPARK Challenge 2025/2026